

## MSc AI Dissertation Project Proposal Form

|                                                                                                                   |                                                      |      |            |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|------|------------|
| Student number                                                                                                    | 202441750                                            | Date | 19/02/2026 |
| Student name                                                                                                      | Dipanjan Das                                         |      |            |
| Supervisor name                                                                                                   | Dr Muhammad Khalid                                   |      |            |
| Project title:                                                                                                    | Sustainable Crop Yield prediction via Multi-modal AI |      |            |
| Project description – background, aims, objectives, research question(s), brief reflection on novelty/originality |                                                      |      |            |

**Background:**

Accurate crop yield forecasting is a critical component of global food security, climate adaptation, and agricultural sustainability. Modern AI systems have improved predictive performance using satellite imagery and meteorological data. However, most state-of-the-art models remain unimodal or naively fused, making them fragile under real-world conditions such as cloud occlusion, missing weather readings, or sensor noise.

Optical satellite systems such as European Space Agency Sentinel-2 are frequently obstructed by cloud cover. Conversely, weather-only models lack spatial granularity and fail to capture localized crop stress. In practice, agricultural data is incomplete, noisy, and temporally misaligned—particularly in developing regions.

This project addresses a fundamental limitation in current multimodal AI which is lack of robustness under modality degradation

**Aim:**

To develop and rigorously evaluate a Gated Multimodal Deep Learning architecture that dynamically adapts to missing or corrupted input modalities for crop yield prediction.

**Objectives:**

- Design unimodal baselines (satellite-only, weather-only).
- Implement early-fusion and late-fusion multimodal baselines.
- Develop a Gated Multimodal Unit (GMU) that dynamically weights modalities.
- Introduce controlled modality dropout to simulate real-world sensor failure.
- Quantify robustness under increasing degradation (10%, 25%, 50%, 75%).
- Compare generalization across geographic regions.

**Novelty and Contribution:**

The novelty lies not in multimodality alone, but in:

- Explicit modelling of modality reliability
- Controlled robustness stress-testing
- Benchmarking resilience rather than ideal-case accuracy
- Architectural bias toward real-world agricultural deployment

This shifts evaluation from best case RMSE to resilience-aware AI design, which remains underexplored in agricultural ML research.

**Research Questions:**

- 1) What is the quantifiable performance of state-of-the-art Unimodal Architectures (specifically 3D-CNN for Satellite-only vs. Bi-LSTM for Weather-only) on any crop related datasets, and under what specific environmental conditions (e.g., >20% cloud cover or extreme drought intervals) do these single-stream models statistically break down?
- 2) Does a gated multimodal architecture significantly reduce predictive performance degradation under simulated modality dropout compared to naïve fusion approaches?
- 3) Can an automated quality check mechanism allow the system to maintain accurate predictions even if 50% of the satellite images are unusable due to cloud cover?

- 4) To what extent does the implementation of a Gated Multimodal Unit (GMU) reduce error variance (RMSE) compared to both Unimodal baselines and standard Static Fusion baselines when subject to non-stationary noise (e.g., synthetic sensor occlusion), thereby establishing a new benchmark for operational resilience?
- 5) Is deep feature-level fusion superior to raw data-level fusion for spatiotemporal agricultural prediction?
- 6) Can a Dual-Stream architecture with intermediate gating autonomously learn to separate data quality from feature importance (i.e., inversely correlating attention weights with pixel-wise entropy/cloud cover) without explicit supervision, providing the interpretability missing in current agricultural AI systems?
- 7) Is it more effective to combine the high-level feature representations from each data source rather than just mixing the raw numbers together?

**Methodology – rationale, data selection and collection, recruitment of participants, analytical process**

**Rationale:** This methodology mimics a human expert agronomist. An expert doesn't just look at a field (Satellite) or just read a thermometer (Weather), they synthesize both. If they can't see the field due to fog, they rely on their knowledge of the temperature. This project encodes that resilience into a mathematical model.

**Data Selection and Collection:** The project uses the CropNet Benchmark (2023), a standard public collection of agricultural data:-

- Input A (Visual): Satellite images (Sentinel-2) that show how green/healthy the plants are over time.
- Input B (Environmental): Daily weather records (Temperature, Rainfall, Sun).
- Target: Historical records of how much crop was actually harvested (Yield).

**Analytical Process:**

- 1) Alignment: Match the daily weather records with the weekly satellite images so they cover the exact same time periods.
- 2) Pattern Recognition: Use software to identify growth trends in the images and stress factors (like drought) in the weather data.
- 3) Smart Fusion: Implement the smart reliability filter to weigh the importance of each source. If it sees clouds, it trusts the weather more. If the weather data is missing, it trusts the images more.
- 4) Stress Testing: Intentionally delete parts of the data (simulating sensor failure) to see if the system can still make good predictions.

**Experimental Design:**

**Baseline model to measure performance against:**

1. CNN (Satellite-only)
2. LSTM (Weather-only)
3. Early Fusion (concatenated raw inputs)
4. Late Fusion (separate encoders + MLP)

**Proposed model:** 3D-CNN and LSTM/Transformer combined with a Gated Multimodal Unit (GMU) that dynamically adjusts modality importance per sample.

**Robustness Testing**

Simulated degradation scenarios:

| Scenario           | Description               |
|--------------------|---------------------------|
| Image dropout      | Random cloud masking      |
| Weather dropout    | Random feature masking    |
| Structured dropout | Seasonal data removal     |
| Combined dropout   | Multi-modality corruption |

Performance degradation curves will be analyzed.

**Evaluation Metrics**

- RMSE
- MAE
- $R^2$
- MAPE
- Relative Degradation Index (RDI)
- Robustness Retention Score (RRS)

**Statistical validation:**

- Paired t-tests
- 5-fold cross-validation
- Geographic holdout validation

**Theoretical framework and intended tools/technologies**

**AI Paradigms:**

- Pattern Recognition: Using standard computer vision (CNNs) to analyze the video of crop growth.
- Time-Series Analysis: Using forecasting methods (LSTMs) to understand weather history.
- Attention Mechanisms: The logic that allows the system to focus on the most important data points.

**Tools/Technologies:**

Software: Python with Jupyter Notebook or Google collab.

Frameworks:

- PyTorch (for building the model structure)
- Numpy, pandas, sklearn, scipy for data preprocessing
- Matplotlib, Seaborn for plots
- Geospatial libraries (for handling map data).

Environment: A anaconda environment with a good graphics card (GPU) for processing the data and training or Google collab

**Key literature identified for literature review (include references)**

Early crop yield prediction research largely relied on single-modality inputs, either remote sensing or climate variables.

Work by Johnson, M. D. & Hsieh, W. W. (2016) demonstrated that vegetation indices such as NDVI derived from satellite imagery can effectively forecast yield using machine learning. However, their framework relies exclusively on optical reflectance signals, limiting the model's ability to capture non-visual stress factors and environmental variability.

Similarly, Schlenker, W. & Roberts, M. J. (2009) showed that temperature thresholds strongly influence crop productivity. Their work established the importance of climatic drivers but focused primarily on meteorological variables without integrating plant-level spatial observations.

Together, these studies show that unimodal systems are informative but incomplete. Each captures only part of the yield formation process.

Lin, F., et al. (2024) introduces CropNet: A Large-Scale Crop Yield Prediction Benchmark. It provides the state-of-the-art benchmark giving spatiotemporal satellite tensors with meteorological data across diverse geographies than any single-source baseline helps in crop yield prediction for deep learning networks.

More recent research, such as Maimaitijiang, M. et al. (2020), empirically demonstrated that combining spectral and thermal UAV data improves yield prediction accuracy compared to single-sensor approaches. This provides direct evidence that multimodal fusion enhances predictive robustness.

From a methodological standpoint, Han, K. et al. (2022) introduced adaptive gating mechanisms for multimodal learning, showing that models can dynamically weigh or suppress noisy modalities. This is particularly relevant for agricultural systems where data sources (e.g., satellite imagery under cloud cover) can be unreliable.

#### References:

- 1) Johnson, M. D., & Hsieh, W. W. (2016). Crop yield forecasting on the Canadian Prairies by remotely sensed vegetation indices and machine learning methods (<https://doi.org/10.1016/j.agrformet.2015.11.003> )
- 2) Schlenker, W., & Roberts, M. J. (2009). Nonlinear temperature effects indicate severe damages to U.S. crop yields under climate change. PNAS.(  
<https://www.pnas.org/doi/epdf/10.1073/pnas.0906865106>)
- 3) Maimaitijiang, M., et al. (2020). Soybean yield prediction from UAV using multimodal data fusion and deep learning.(<https://doi.org/10.1016/j.rse.2019.111599> )
- 4) Lin, F., et al. (2024). An Open and Large-Scale Dataset for Multi-Modal Climate Change-aware Crop Yield Predictions (<https://arxiv.org/pdf/2406.06081> )
- 5) Han, Z., et al. (2022). Multimodal Dynamics: Dynamical Fusion for Trustworthy Multimodal Classification ([https://openaccess.thecvf.com/content/CVPR2022/html/Han\\_Multimodal\\_Dynamics\\_Dynamical\\_Fusion\\_for\\_Trustworthy\\_Multimodal\\_Classification\\_CVPR\\_2022\\_paper.html](https://openaccess.thecvf.com/content/CVPR2022/html/Han_Multimodal_Dynamics_Dynamical_Fusion_for_Trustworthy_Multimodal_Classification_CVPR_2022_paper.html) )

**Industrial partner (if appropriate)**

N/A

**Project management**

Table: Project timeline and key outputs (expand table as needed)

| <b>Week</b>  | <b>Activity</b>                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 5       | Submit Final Project Proposal                                                                                                                         |
| Week 6       | Draft Ethical Report Submission and review with supervisor                                                                                            |
| Week 7       | Final Ethical Report Submission                                                                                                                       |
| Week 8       | Interim Report discussion with supervisor                                                                                                             |
| Week 10 - 11 | Initial EDA analysis discussion and model architecture discussion with supervisor                                                                     |
| Week 13      | Interim Report submission                                                                                                                             |
| Week 14      | Get feedback for interim report and Final EDA analysis discussion with supervisor                                                                     |
| Week 15      | Developing a model for handling tabular data and doing hyper parameter tuning and review discussion with supervisor                                   |
| Week 16      | Developing a model for handling imagery and review discussion with supervisor                                                                         |
| Week 17      | Developing a gating logic for both the model and doing hyper parameter tuning and calculating important metrics and review discussion with supervisor |
| Week 19      | Doing stress testing and calculating and comparing benchmarks with other models and reviewing it with supervisor                                      |
| Week 21      | Initial Project Report and Presentation discussion with supervisor                                                                                    |
| Week 22      | Initial Project Report and Presentation review and feedback with supervisor                                                                           |
| Week 23      | Final review of Project Report, Presentation and Portfolio with supervisor                                                                            |

**Resources and Research Data Management Plan (Describe the resources required and the data you expect to acquire or generate during this research project, how you will manage, describe, analyse, and store the data and what mechanisms you will use to share and preserve your data.)**

#### Resources:-

- PyTorch Documentation (<https://docs.pytorch.org/docs/stable/index.html>)
- RasterIO/Geopandas (<https://rasterio.readthedocs.io/en/latest/> , <https://geopandas.org/en/stable/docs.html> )
- Scikit-learn (Analysis) ([https://scikit-learn.org/stable/user\\_guide.html](https://scikit-learn.org/stable/user_guide.html) )
- CropNet Dataset Documentation (<https://huggingface.co/datasets/CropNet/CropNet> )

#### Research Data Management:-

1. Data Profiling and Multi-modal Alignment - Before diving into individual features, we have to verify how the three modalities (Sentinel-2, WRF-HRRR, and USDA) align.

- Coordinate Verification: Ensure that the 224x224 RGB/NDVI images match the 9x9 km meteorological grids and the specific US county they are assigned to.
- Temporal Resolution Check: The Sentinel-2 images are updated every 14 days, WRF-HRRR is daily/monthly, and USDA yields are annual. Check for missing time steps across these different scales.

#### 2. Modality-Specific EDA

##### A. Sentinel-2 Satellite Imagery (Visual/Spectral)

- NDVI Distribution: Plot the distribution of Normalized Difference Vegetation Index (NDVI) values. Healthy vegetation typically scores between 0.3 and 0.8.
- Seasonality Visualization: Select a few specific counties and plot their image sequence over 12 months to observe the greening up and browning down phases.
- Outlier Detection: Look for images with heavy cloud cover or missing pixels (No-Data values), which can skew yield predictions.

##### B. WRF-HRRR Meteorological Data (Numerical)

- Time-Series Analysis: Plot parameters like Averaged Temperature and Precipitation over the growing season. Use rolling averages to smooth out daily fluctuations.
- Correlation Heatmap: Check the relationship between weather variables. For example, high temperatures often correlate with low relative humidity.
- Stationarity Tests: Since we are likely using these for time-series forecasting, perform Augmented Dickey-Fuller (ADF) tests to check for trends or seasonality.

##### C. USDA Crop Dataset (Tabular Labels)

- Target Variable Distribution: Use a histogram to visualize the distribution of Crop Yield (tons per hectare). Agricultural data often has a long tail or is skewed by exceptionally good or bad harvest years.
- Geographic Visualization: Create a choropleth map of the U.S. to see which counties have the highest yield density for specific crops (Corn, Cotton, Soybeans, Winter Wheat).

#### 3. Cross-Modality Relationship Analysis

This is the most critical step for project.



- Weather vs. Yield: Scatter plots of "Cumulative Growing Degree Days" (calculated from temperature) against the final annual yield.
- NDVI vs. Yield: Perform a Pearson correlation between the peak NDVI value reached in the summer and the year-end harvest weight.
- Lag Analysis: Analyze how weather events in the spring (e.g., heavy rainfall) impact the vegetation health (NDVI) measured weeks later in the summer.

#### 4. Feature Engineering Exploration

Based on EDA, we will identify the need to create new features:

- Anomaly Scores: Calculate the deviation of current year weather from the 3-year county average.
- Crop Phenology Markers: Identify the exact date when NDVI reaches its maximum for each county, which can serve as a powerful feature for yield timing.

#### Planned outputs/publications/research datasets/impact/dissemination

- Primary Output: Establishes a benchmark for noise robust architecture design in geospatial deep learning, demonstrating how architectural priors (gating) can substitute for data cleaning in resource-constrained environments.
- Primary Output: A technical analysis of the loss landscape and generalization bounds of Gated Multimodal Networks.
- Secondary Output: A possible modular PyTorch implementation of the GMU architecture, released as an open-source library.

If successful, I undertake to carry out the research according to the University's Ethics code of Practice. I realise that I will not proceed into data collection without an ethical approval in place

(Applicant's signature required)

Dipanjan Das

Date and signature of Supervisor approval